

THEORY AND METHODS

**MODELLING SPECIES-ABUNDANCE
FREQUENCIES AND SPECIES-INDIVIDUAL
FUNCTIONS WITH THE GENERALIZED
INVERSE GAUSSIAN-POISSON DISTRIBUTION**

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for publication in this volume was carried out by Paul Fatti.

Key words: Generalized inverse Gaussian-Poisson distribution; species-abundance distribution; species-individual function.

Summary: The generalized inverse Gaussian-Poisson distribution (GIGP) is suggested as a new stochastic model for species-abundance distributions and for species-individual functions. It has many theoretical and practical advantages over previously used mathematical models. A number of applications show good agreement with observed data.

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1. Introduction

Ever since Fisher *et al.* (1943) derived the logarithmic series distribution for species frequency, a lively discussion has taken place about the suitability of that model. Fisher postulated that, for a given ecological population in a well defined area during equal time periods, the number of individuals of a given species would be distributed in accordance with Poisson's probability law for $r = 0, 1, 2, \dots$ with parameter λ_i . Different species had different λ_i 's which

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themselves would be distributed in the total population of S^* species according to the gamma distribution function:

$$f(\lambda) = \frac{1}{\Gamma(\gamma)} \left(\frac{1-\theta}{\theta} \right)^\gamma \lambda^{\gamma-1} e^{-(\frac{1-\theta}{\theta})\lambda}, \quad (1)$$

where $\lambda \geq 0$, $\gamma > 0$ and $0 < \theta < 1$.

Note that this 'mixing' distribution does not exist for $\gamma \leq 0$. The convolution of the Poisson with the gamma distribution led to the well-known negative binomial (NBD)

$$\begin{aligned} \phi(r) &= \frac{(1-\theta)^\gamma \Gamma(r+\gamma)}{\Gamma(\gamma) r!} \theta^r \\ &= (1-\theta)^\gamma \binom{-\gamma}{r} (-\theta)^r, \quad \text{for } r = 0, 1, 2, \dots \end{aligned} \quad (2)$$

As the number of species not represented in a collection was unknown and unobservable, the NBD in (2) had to be zero-truncated giving

$$\phi(r) = [(1-\theta)^{-\gamma} - 1]^{-1} \binom{-\gamma}{r} (-\theta)^r, \quad \text{for } r = 1, 2, 3, \dots \quad (3)$$

Fisher noted that in estimating the parameters of the zero-truncated NBD from real data, $\hat{\gamma}$ (in his notation $\hat{k}$) was small **and** positive. He then took the controversial step of letting $\gamma \rightarrow 0$ which resulted in

$$\lim_{\gamma \rightarrow 0} \phi(r) = \psi(r) = [-\ln(1-\theta)]^{-1} \frac{\theta^r}{r}, \quad \text{for } r = 1, 2, 3, \dots \quad (4)$$

Fisher's logarithmic series distribution (LSD) in (4) cannot be extrapolated to $\psi(0)$ and hence it is impossible to estimate the total number of species S^* in the population. As a matter of fact, $\hat{\psi}(0) = \infty$ and hence $\hat{S}^* = \infty$ which is in

stark contrast to the known ecological fact that all species populations are finite. Of course, Fisher stated distinctly in his paper that $\gamma \neq 0$ and that S^* was finite. He considered (4) as a reasonable simplified approximation to the more correct zero-truncated NBD, if γ was small and positive. In addition, some observed species frequency distributions were satisfactorily represented by the LSD at the time when Fisher published his paper.

From the zero-truncated NBD in (3) it is, however, possible to estimate $\phi(0)$ and hence S^* (Pielou, 1969). Why then was the zero-truncated NBD not used? The answer to this question was put forward by Kempton and Taylor (1974) who showed that the maximum likelihood support function for S^* and γ gave rise to an ill-defined maximum support so that it appeared to be useless to estimate the population size S^* .

Whilst this argument is undoubtedly correct, it is unfortunate that it leads to an approximation which appears to be nonsensical as it encompasses a false infinite population concept. It appears much better to acknowledge that the estimate of S^* has very wide confidence limits than to deny its finite existence altogether.

There are two further shortcomings of the LSD model.

1. The Fisher distribution cannot reproduce a modal frequency, except at $r=1$.

There are however, observed species frequency distributions in existence where the mode is situated at $r = 2$ or 3. In any case, given a finite population size S^* and a very long time period over which data are collected, then it stands to reason that the mode must eventually shift to $r > 1$.

2. Kempton and Taylor (1974) and Taylor *et al.* (1976) have shown that many

observed species frequency distributions are fitted badly by the zero-truncated NBD or by the LSD in as much as the tails of the observed distributions are much longer than what is allowed for by these two models.

As an alternative, other authors such as Grundy (1951), Bulmer (1974) and also Kempton and Taylor (1974) made use of the discrete Poisson-lognormal distribution. Although this model has the great advantage of allowing for an estimate of the total number of species in the population S^* , it is mathematically awkward and recourse must be made to a vast amount of numerical integration. One of its real disadvantages is its excessive skewness so that the observed tail now falls short of the theoretical one which is the opposite to what was said about the log-series (Kempton and Taylor, 1974).

Engen (1978) suggested another model for species frequencies which he called the extended negative binomial. By reversing the sign of the shape parameter γ in (3) one obtains

$$\phi(r) = [(1 - \theta)^\gamma - 1]^{-1} \binom{\gamma}{r} (-\theta)^r, \quad \text{for } r = 1, 2, 3, \dots \quad (5)$$

This then is a zero-truncated NBD with a **negative** index of the shape parameter, *i.e.* $-\infty < \gamma < 0$ and $0 < \theta < 1$. As $\gamma \rightarrow 0$, we obtain once again Fisher's LSD. For computational purposes γ is taken as positive in (5). This distribution fits a good number of observed species frequencies rather well as its skewness lies somewhere between that of the log-series and the Poisson-lognormal.

Its disadvantages are similar to those found in the log-series: One cannot estimate the population size S^* thus introducing a conceptual error in the

mathematical model. Further, it is incapable of reproducing a mode except at $r = 1$.

Although more flexible than Fisher's distribution, the generalized log-series (Kempton, 1975) suffers of the same structural shortcomings as the ordinary LSD.

2. A new model for the species-abundance distribution

Under the conditions as previously stated, the individuals of a given single species are assumed to be Poisson distributed with an expected average number of λ per unit time and $t\lambda$ in a time period of length t . The convolution integral becomes

$$\phi(r|t) = \int_0^\infty \frac{e^{-t\lambda}(t\lambda)^r}{r!} f(\lambda) d\lambda, \quad \text{for } r = 0, 1, 2, \dots \text{ and } t > 0. \quad (6)$$

In repeated periods of equal length t the total number of individuals observed for all species will be a random variable $\hat{N}$ with an expectation of $E(\hat{N}) = N$.

As N is proportional to t in the long run, we may replace t by N in (6) giving

$$\phi(r|N) = \int_0^\infty \frac{e^{-N\lambda}(N\lambda)^r}{r!} f(\lambda) d\lambda, \quad \text{for } r = 0, 1, 2, \dots \quad (7)$$

For the mixing distribution we use the generalized inverse Gaussian (GIG)

$$f(\lambda) = \frac{(2/bc)^\gamma}{2K_\gamma(b)} \lambda^{\gamma-1} e^{-\frac{\lambda}{c} - \frac{b^2 c}{4\lambda}} \quad \text{for } \lambda \geq 0, \quad (1)$$

$-\infty < \gamma < \infty$, $b \geq 0$ and $c > 0$.

$K_\gamma(b)$ is the modified Bessel function of the second kind of order γ and argument b .

For $b = 0$ the GIG is a gamma (Pearson Type III) distribution and for $c \rightarrow \infty$ it is a Pearson Type V. This indicates that the generalized inverse Gaussian distribution is extremely flexible and much richer than the customary gamma probability function.

A notable feature of the GIG distribution in (8) is its existence even if the shape parameter γ extends over the very wide range of $(-\infty, \infty)$.

Consequently, we should not expect any difficulty with $\gamma = 0$ or with negative γ 's, as previously experienced with the gamma mixing distribution.

Further, the GIG distribution **always** has a mode. For certain combinations of its parameters the maximum is very near to $\lambda = 0$ so that in such cases its appearance is that of a reverse *J*-shaped distribution.

The convolution integral in (7), with $f(\lambda)$ as in (8), was first solved by the present author (Sichel, 1971).

In the current notation (Sichel, 1986)

$$\phi(r|N) = \left[(1 + cN)^{\gamma/2} K_{\gamma}(b) \right]^{-1} \frac{1}{r!} \left(\frac{bcN}{2\sqrt{1 + cN}} \right)^r \quad (9)$$

$$K_{r+\gamma}(b\sqrt{1 + cN}), \quad \text{for } r = 0, 1, 2, \dots$$

This is the generalized inverse Gaussian-Poisson (GIGP) distribution.

As species with $r = 0$ individuals cannot be observed, we have to zero-truncate the distribution in (9). In addition we reparameterize by writing

$$\alpha_N = b\sqrt{1 + cN}, \quad (10)$$

where $\alpha_N \geq 0$. This new parameter α_N is functionally dependent on the total number of individuals N , in contrast to the invariant parameters γ , b and c

which were inherited from the mixing distribution in (8). From (10) we also obtain

$$c = \frac{1}{N} [(\alpha_N/b)^2 - 1]. \quad (11)$$

Substitution of (10) and (11) into (9), and after zero-truncation, we finally have

$$\phi(r|N) = [(\alpha_N/b)^\gamma K_\gamma(b) - K_\gamma(\alpha_N)]^{-1} \frac{[(\alpha_N^2 - b^2)/2\alpha_N]^r}{r!} K_{r+\gamma}(\alpha_N), \quad \text{for } r = 1, 2, 3, \dots \quad (12)$$

This is the zero-truncated GIGP distribution.

This distribution is quite flexible and it can be shown (Sichel, 1992) that for

(i) $\alpha_N \rightarrow 0$, $b \rightarrow 0$ and $\gamma > 0$, the distribution becomes the traditional zero-truncated negative binomial as in (3), if one writes $\theta_N = 1 - (b/\alpha_N)^2$.

(ii) $\alpha_N \rightarrow 0$, $b \rightarrow 0$ and $\gamma = 0$, the distribution becomes Fisher's LSD as in (4).

(iii) $\alpha_N \rightarrow 0$, $b \rightarrow 0$ and $\gamma < 0$, the distribution becomes Engen's extended negative binomial as in (5).

In general it is possible to estimate the frequency for the unseen species (at 0) from the zero-truncated GIGP distribution in equation (12) by

$$\text{Freq}(0) = S_N \left[(\alpha_N/b)^\gamma \frac{K_\gamma(b)}{K_\gamma(\alpha_N)} - 1 \right]^{-1}, \quad (13)$$

where S_N is the expected number of species in repeated samples of N individuals. From (13) the total number of species in the population is

$$S^* = S_N + \text{Freq}(0). \quad (14)$$

As previously stated, one **cannot** estimate S^* from the traditional Fisher's LSD or from Engen's extended NBD.

The last mentioned two distributions are always reverse J -shaped and hence they cannot represent observed species-abundance distributions with a mode at $r > 1$.

On the other hand, the GIGP distribution in (12) can be reverse J -shaped or unimodal with a mode at $r \geq 1$.

As the general compound Poisson distribution in (7) inherits its shape from the mixing distribution $f(\lambda)$, we determined the skewness for the gamma, the generalized inverse Gaussian and the lognormal mixing distributions with the first two moments made all the same. In other words the coefficients of variation (CV) for all three mixing distributions were identical.

We found:

$$\text{Gamma: } \sqrt{\beta_1} = 2CV,$$

$$\text{Generalized inverse Gaussian: } \sqrt{\beta_1} \approx \left(\frac{2+\gamma}{1+\gamma}\right)CV,$$

$$\text{Lognormal: } \sqrt{\beta_1} = 3CV[1 + \frac{1}{3}CV^2].$$

A γ value of $-\frac{1}{2}$ for the GIG distribution makes $\sqrt{\beta_1} = 3CV$. (For $\gamma = -\frac{1}{2}$ this is an exact formula.)

Hence it follows that the skewness of the generalized inverse Gaussian distribution is somewhere between the skewness of the gamma and the

lognormal. It will be remembered that according to Taylor *et al.* (1976) and Kempton and Taylor (1974) observed species-abundance distributions often had tails longer than the log-series but shorter than the Poisson-lognormal distribution.

3. Parameter estimation for the species-abundance distribution

For the joint estimation of the three parameters α_N , b and γ of the GIGP distribution in (12) we employ the "constrained minimum χ^2 " method. The constraint arises from the constancy of the observed arithmetic mean $\hat{E}(r|N) = N/\hat{S}_N$ which will not be changed through any operation. The arithmetic mean of the GIGP in (12) is a Maximum Likelihood estimator. The population mean of the zero-truncated GIGP in (12) is (Sichel, 1986)

$$E(r|N) = \frac{[(\alpha_N^2 - b^2)/2b]K_{\gamma+1}(b)}{K_{\gamma}(b) - (b/\alpha_N)^{\gamma}K_{\gamma}(\alpha_N)} = N/S_N \quad (15)$$

and the population proportion of singletons is from (12)

$$\phi(1|N) = \frac{[(\alpha_N^2 - b^2)/2\alpha_N]K_{\gamma+1}(\alpha_N)}{(\alpha_N/b)^{\gamma}K_{\gamma}(b) - K_{\gamma}(\alpha_N)} = \text{Freq}(1|N)/S_N. \quad (16)$$

Division of (16) by (15) leads to

$$\text{Freq}(1|N) = N(b/\alpha_N)^{\gamma+1} \frac{K_{\gamma+1}(\alpha_N)}{K_{\gamma+1}(b)}. \quad (17)$$

We start off with an initial $\hat{\gamma}_0$ (say -0.25) and an initial $\hat{\alpha}_{N,0}$ and we solve (17)

for b , with the Newton-Raphson method. For $\text{Freq}(1|N)$ in (17) we initially

use the observed $\widehat{\text{Freq}}(1|N)$.

Next we substitute estimates $\hat{\gamma}_0$, $\hat{\alpha}_{N,0}$ and $\hat{b}_0$ into (16). This will in general not match the observed $\hat{\phi}(1|N) = \widehat{\text{Freq}}(1|N)/\hat{S}_N$. We now change $\hat{\alpha}_N$ in (17) systematically, retain $\hat{\gamma}_0$ and solve for $\hat{b}$ until we have found a pair $(\hat{\alpha}_N, \hat{b}|\hat{\gamma}_0)$ which, after substitution into (16), matches the observed $\hat{\phi}(1|N)$.

We now calculate the χ^2 -statistic of goodness-of-fit. Repeating this process for different $\hat{\gamma}'$ s systematically, we arrive at a minimum χ^2 solution which has the expected $\widehat{\text{Freq}}(1|N)$ matching the observed $\widehat{\text{Freq}}(1|N)$ exactly. The observed and expected means $\widehat{E}(r|N)$ and $E(r|N)$ will also be the same exactly.

We now carry on the same algorithm but changing the $\widehat{\text{Freq}}(1|N)$ in (17) systematically.

Eventually we find a **global** minimum χ^2 where the observed and expected arithmetic means still correspond exactly, but where the observed and expected singleton frequencies differ slightly from each other.

It should be noted that for large sample sizes $\hat{S}_N$, maximum likelihood and minimum χ^2 parameter estimates are virtually identical according to theory.

In any case, even if we had used a maximum likelihood estimation method, we still would have to calculate the χ^2 -statistic to test whether the chosen GIGP model was suitable.

4. A new model for the species-individual function

The species-individual function should pass through the origin and should be approximately linear for small numbers of individuals N , with a slope of unity. Further, it should reach a finite upper asymptote for large numbers of individuals N as the total number of observed species S^* in ecological

populations is always finite. This desideratum automatically excludes Fisher's LSD and Engen's extended NBD as both models have $S^* \rightarrow \infty$.

The traditional NBD, the Poisson-lognormal and the GIGP do have finite upper asymptotes and we have to judge via data analysis which of these three models is the best to apply in reality.

It has become standard practice to plot the logarithms of the observed number of individuals $\hat{N}$ on the abscissa and the observed number of species $\hat{S}_N$ on the ordinate when displaying an observed data set graphically. In that case the representative population curve should look like a cumulative distribution function with a lower asymptote near $S_1 = 1$, rising to a straight line portion for medium size N and eventually reaching an upper asymptote at $S_\infty = S^*$. As most data sets start at comparatively large N and as they are rarely observed at very large N , the resulting plotted points look like a linear function. This has led writers to the belief that mathematical models like the log-series or equations like $y = ax^b$ are appropriate. This assumption is however erroneous from a theoretical point of view, as it totally neglects the finite nature of a species-abundance population.

The three major portions of the species-log-individual curves are well illustrated by a set of real data (Taylor *et al.*, 1976, p.266, Fig.3). The species-individual curve in its most general form is

$$S_N = S^*[1 - \phi(0|N)] \quad (18)$$

where $\phi(0|N)$ is the probability that species which belong to the population of S^* are not observed in a count of N individuals (Sichel, 1986). From (9) we

$$\phi(0|N) = \frac{K_\gamma(b\sqrt{1+cN})}{(\sqrt{1+cN})^\gamma K_\gamma(b)} \quad (19)$$

Further, it was shown (Sichel, 1986) that generally

$$S^* = [E(\lambda)]^{-1} \quad (20)$$

which is the inverse of the mean of the mixing distribution $f(\lambda)$.

From the GIG distribution in (8) we find

$$E(\lambda) = \frac{1}{2} bc \frac{K_{\gamma+1}(b)}{K_\gamma(b)} \quad (21)$$

From (18), (19), (20) and (21) we obtain finally

$$S_N = \frac{2}{bc} \frac{K_\gamma(b)}{K_{\gamma+1}(b)} \left[1 - \frac{K_\gamma(b\sqrt{1+cN})}{(\sqrt{1+cN})^\gamma K_\gamma(b)} \right] \quad (22)$$

This is the species-individual function S_N with invariant parameters b , c and γ and a count of N individuals.

It should be emphasized that S_N is **not** an empirical function. It is based on sound probability theory.

5. Parameter estimation for the species-individual function

The function in (22) has three invariant parameters b , c and γ , all of which are important with respect to the shape of the species-individual or species-log-individual curves. From the pairs of observations $(N, \hat{S}_N)$ the species-individual curve is fitted by a non-linear unweighted least square method.

which gives 'best' joint estimates $\hat{b}$, $\hat{c}$ and $\hat{\gamma}$ simultaneously. The mathematical technique and the computer program were developed recently (Price, 1991). They appear elsewhere in this volume (Price, 1997).

6. Applications

(a) Species-Abundance Distributions

The first example refers to the observed distribution of extant rare plant species in the southern Appalachians (Miller and Wiegert, 1989). There were $N = 1008$ individuals in the collection with $\hat{S}_N = 188$ distinct species. The number of observed singleton species was $\text{Freq}(1|N) = 61$ compared to the statistical estimate of $\widehat{\text{Freq}}(1|N) = 59.5$. The parameter estimates via the constrained minimum χ^2 method were:

$$\hat{\gamma} = -0.370,$$

$$\hat{\alpha}_N = 1.201792,$$

$$\hat{b} = 0.247308,$$

$$\hat{\theta}_N = 1 - (\hat{b}/\hat{\alpha}_N)^2 = 0.9576535.$$

The minimum χ^2 was 8.358 with 10 degrees of freedom and an associated probability of $P(\chi^2) = 0.594$. The estimate for the frequency of unobserved species was $\widehat{\text{Freq}}(0|N) = 95.317$ giving an estimate for the total number of species in the population of $\hat{S}^* = 283.317$.

Observed and expected species frequencies are shown in table 1.

Table 1: Observed and expected species-abundance distribution of extant rare plants in Southern Appalachians [Miller and Wiegert, 1989].

Number of individuals per species	Observed number of species	Expected number of species
1	61	59.500
2	35	33.731
3	18	20.835
4	12	14.050
5	15	10.113
6	4	7.628
7	8	5.955
8	4	4.772
9	5	3.901
10	5	3.242
11-14	6	8.779
15-19	6	5.950
20-29	4	5.389
≥ 30	5	4.155
$\hat{S}_N$	188	188.000
$\hat{\gamma}$		-0.37000
$\hat{a}_N$		1.20179
$\hat{b}$		0.24731
$\hat{\theta}_N$	$= 1 - (\hat{b}/\hat{a}_N)^2$	0.95765
d.f.		10
χ^2		8.358
$P(\chi^2)$		0.594
S^*		283

One of the best known observed species-abundance distributions in the literature is that of Corbet's Malayan butterflies collection (Bulmer, 1974). Corbet observed 620 distinct species. The distribution was censored giving 119 species for $r \geq 25$. Corbet himself estimated that the total number of species — seen and unseen — was $\hat{S}^* = 924$. Fisher's LSD fitted the data very poorly and the best model obtained so far was the Poisson-lognormal with a χ^2 statistic of goodness-of-fit of 30.527 with 22 degrees of freedom and an associated probability of $P(\chi^2) = 0.106$ (Bulmer, 1974).

Due to the censoring of the upper tail no total number of individuals N was given by Corbet. However, a rough estimate is $\hat{N} \sim 9800$. Further, $\hat{S}_N = 620$ and $\text{Freq}(1|N) = 118$, compared to the statistical estimate of $\text{Freq}(1|N) = 116.4$, based on the GIGP model. The parameter estimates via the constrained minimum χ^2 method were:

$$\hat{\gamma} = 0.140,$$

$$\hat{\alpha}_N = 0.144927,$$

$$\hat{b} = 0.0209063,$$

$$\hat{\theta}_N = 1 - (\hat{b}/\hat{\alpha}_N)^2 = 0.9791908.$$

The minimum χ^2 was 20.077 with 21 degrees of freedom and an associated probability of $P(\chi^2) = 0.516$. This result is far superior to the one achieved by Bulmer, as mentioned above. Hence one must conclude that the GIGP hypothesis is much nearer to the truth than is the Poisson-lognormal model, at least in this particular case.

It is interesting to note that recently Corbet's data were also fitted to the inverse Gaussian-Poisson distribution (Ord and Whitmore, 1986). The IGP is a special case of the GIGP with parameter $\gamma = -\frac{1}{2}$ chosen a priori. Their expected frequencies were disappointing as the goodness-of-fit statistic χ^2 was 44.37 with 22 degrees of freedom. This finding shows clearly how important the choice of parameter γ is. For species-abundance studies one finds $-\frac{1}{2} < \gamma < \frac{1}{2}$. One should estimate γ from the data which means that the GIGP model is much more appropriate than the IGP.

The GIGP fit to the Malayan butterfly observations is shown in table 2.

(b) Species-Individual Functions

Williams has given a good number of pairs of observations ($N, \hat{S}_N$) referring to moths (Macro-lepidoptera) collected in light traps at Rothamsted, England, over many years (Fisher *et al.*, 1943, p. 48; Williams 1953, pp. 20, 24 and 25).

The 32 pairs of observations and the corresponding expected frequencies, based on the species-individual function of equation (22), are shown in table 3.

From the non-linear unweighted least square estimation method we found

$$\hat{b} = 0.00309329,$$

$$\hat{c} = 0.03846803,$$

$$\hat{\gamma} = -0.11584494.$$

The estimate for the total number of species in the population, from the inverse of (21), was

$$\hat{S}^* = \frac{2}{\hat{b}\hat{c}} \frac{K_{\hat{\gamma}}(\hat{b})}{K_{\hat{\gamma}+1}(\hat{b})} = 653.8.$$

Table 2: Observed and expected species-abundance distribution of Corbet's Malayan butterflies. [Bulmer, 1974.]

Number of individuals per species	Observed number of species	Expected number of species
1	118	116.400
2	74	65.923
3	44	46.144
4	24	35.497
5	29	28.792
6	22	24.157
7	20	20.752
8	19	18.138
9	20	16.065
10	15	14.379
11	12	12.980
12	14	11.799
13	6	10.790
14	12	9.917
15	6	9.154
16	9	8.482
17	9	7.885
18	6	7.352
19	10	6.874
20	10	6.441
21	11	6.049
22	5	5.692
23	3	5.365
24	3	5.065
≥ 25	119	119.912
$\hat{S}_N$	620	620.000
$\hat{\gamma}$		0.140000
$\hat{a}_N$		0.144927
$\hat{b}$		0.020906
$\hat{\theta}$	$= 1 - (\hat{b}/\hat{a}_N)^2$	0.979191
d.f.		21
χ^2		20.077
$P(\chi^2)$		0.516
$\hat{S}^*$		1000

Table 3: Observed and expected species-individual function for moths (macro-lepidoptera) at Rothamsted [Williams, 1953].

Number of Individuals N	Number of Species observed $\hat{S}_N$	Number of Species expected $\hat{S}_n$
250	67.6	70.6
409	86.8	86.6
440	89.6	89.1
855	107.1	113.0
1230	119.7	127.2
1961	154	146.3
3124	156	166.6
3276	168	168.7
3454	173	171.1
3771	187	175.1
3786	184	175.3
5252	200	190.6
5517	206	192.9
5972	197	196.8
6107	191	197.8
6530	191	201.1
6815	204	203.2
7378	197	207.1
8792	219	215.8
8794	216	215.9
10092	220	222.9
10357	222	224.2
12069	229	232.1
12334	232	233.2
12456	249	233.8
13632	234	238.5
15221	234	244.4
16972	254	250.3
26300	265	274.5
32853	285	287.3
35428	304	291.7
87400	346	346.6
$\hat{b} = 0.00309329$ $\hat{c} = 0.03846803$ $\hat{\gamma} = -0.11584494$ $\hat{S}^* = 653.8$		

This is a very reasonable estimate as the actual total number of individuals and species found to date in England consisted of "more than two million individuals in over 600 species" (Taylor, 1978, p. 6).

The data of table 3 have been plotted as a species-individual curve in figure 1 and as a species-log-individual curve in figure 2. In the latter graph the smooth theoretical curve covers the whole possible range, with the two expected asymptotes shown as previously discussed. Real data rarely cover the entire range.

The second example for a species-individual function refers to a spread sample of moths taken at different time intervals over an eight-year time horizon at Geescroft Wilderness, England, during the period 1965-72 (Taylor *et al.* 1976).

The data and the resulting fit to equation (22) are shown in table 4.

In figure 3 the species-individual curve for the moths at Geescroft Wilderness has been plotted.

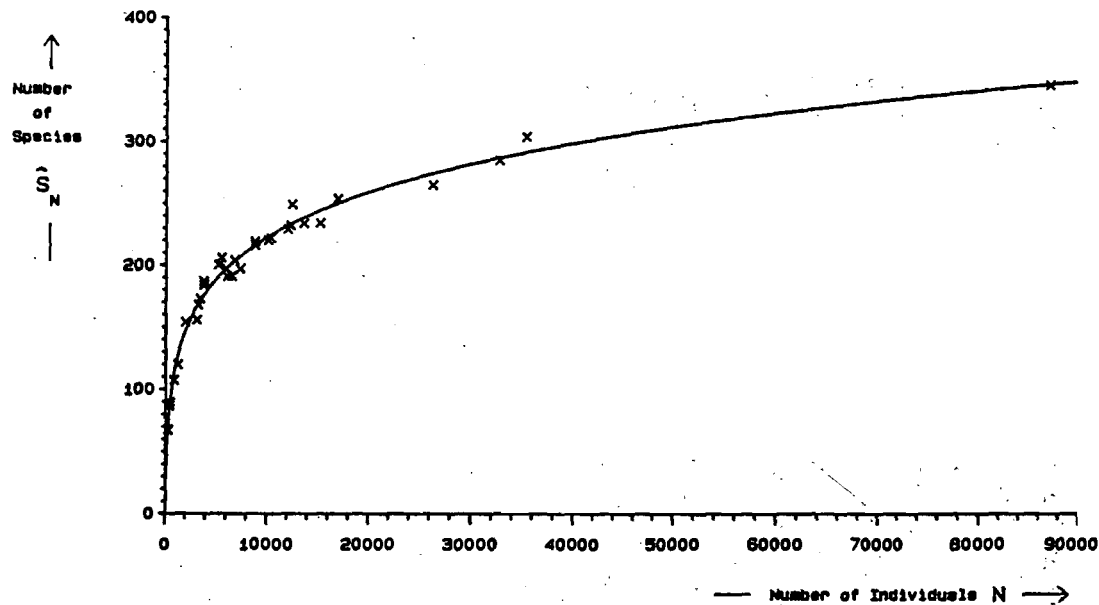


Figure 1. Species-individual curve for macro-lepidoptera at Rothamsted.

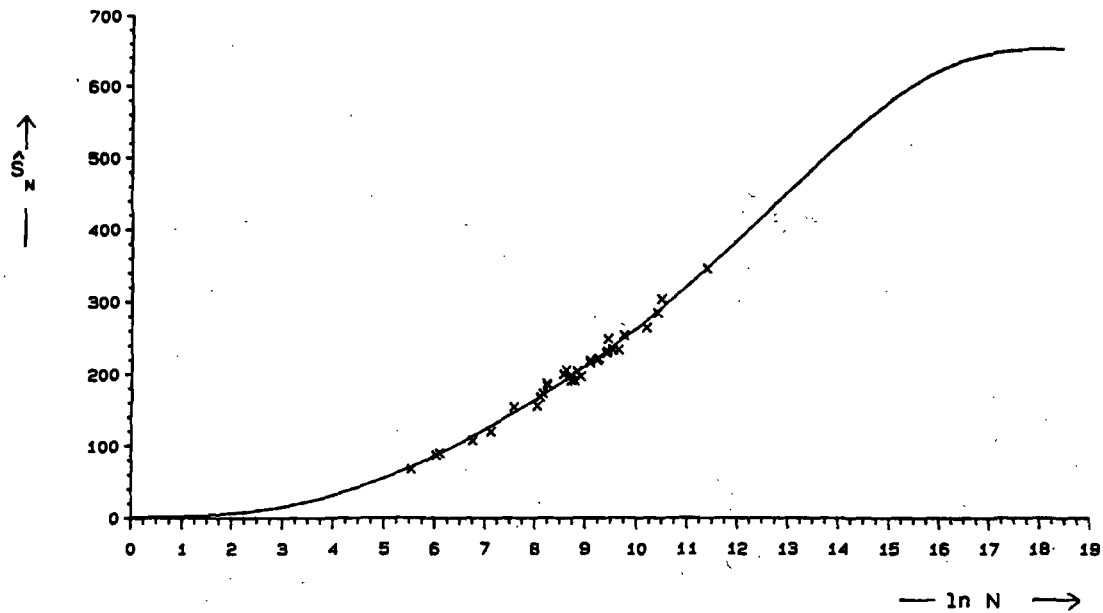


Figure 2. Species-log-individual curve for macro-lepidoptera at Rothamsted.

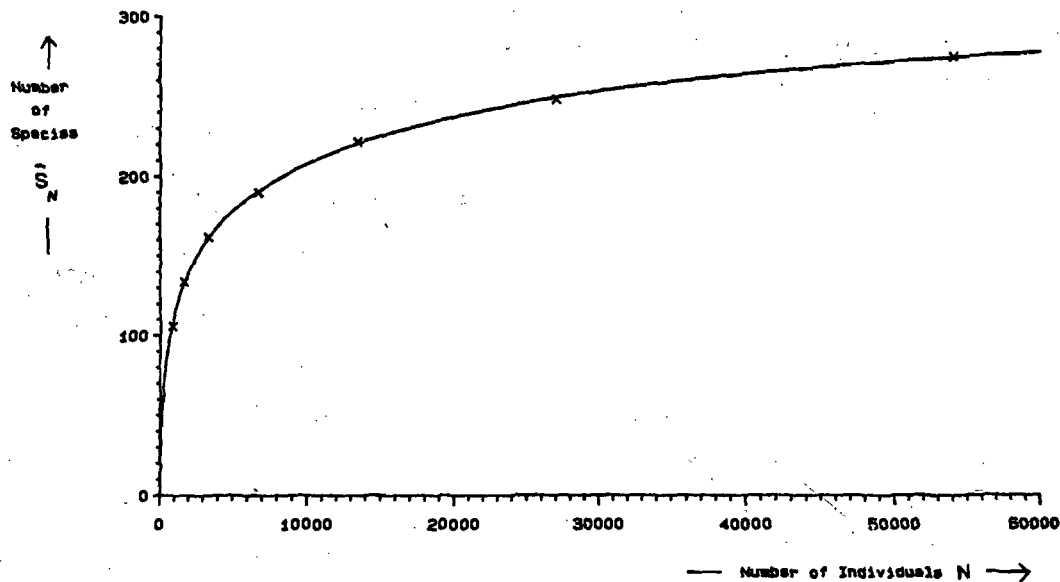


Figure 3. Observed and expected species-individual function for moths at Geescroft Wilderness, England, during 1965-72.

Table 4: Observed and expected species-individual function for moths at Geescroft Wilderness, England, during 1965 to 1972 [Taylor *et al.*, 1976].

Number of Individuals N	Number of Species observed $\hat{S}_N$	Number of Species expected $\tilde{S}_N$
843	106	107.0
1686	134	133.2
3373	162	161.4
6746	190	190.9
13493	222	220.8
26986	248	249.3
53973	275	274.5
$\hat{b}$		0.02490728
$\hat{c}$		0.04516948
$\hat{\gamma}$		-0.15215733
$\hat{S}^*$		317.8

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